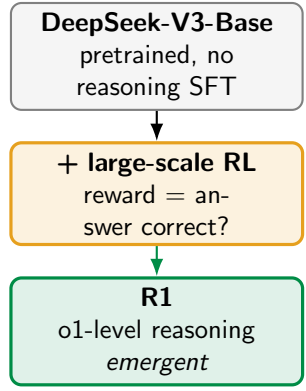
An open model reaches o1-level – and nobody taught it to reason

DeepSeek-R1 matches OpenAI's **o1** on competition math (AIME 79.8 vs 79.2) and coding (Codeforces 96th percentile) – open weights, published recipe.

The striking part is *how* it got there. The reasoning was **not demonstrated in training data**. Starting from a base model and a simple "is the answer correct?" reward, the model **taught itself** to think longer, check its work, and backtrack.

Advanced reasoning **emerged** from reinforcement learning – no human reasoning traces required.

Today: *how pure RL incentivizes reasoning, the full R1 pipeline, and why distilling R1 beats running RL on a small model.*



Outline

The setup

A pretrained base, a reward that only checks the answer, and one RL algorithm.

The starting point and the target

Start from DeepSeek-V3-Base – a strong pretrained model, but *not* fine-tuned to reason.

Target: strong reasoning on *verifiable* tasks – math, code, STEM – where an answer can be checked automatically.

The bet: do **not** start with supervised reasoning traces. The authors hypothesize that imitating human-written chains *caps* the model at human-like reasoning and blocks it from discovering its own.

Instead of teaching *how* to reason (SFT on chains), give the model a goal (*get the answer right*) and let RL find the reasoning that achieves it.

First experiment – **R1-Zero** – is the extreme version: **pure RL, zero SFT**.

The reward is rule-based, not a neural model

For verifiable tasks, the reward is computed by **rules**, not a learned network:

$$R_{\text{rule}} = R_{\text{accuracy}} + R_{\text{format}}$$

- **Accuracy:** is the final answer correct? Math answers go in a box and are checked; code is run against test cases.
- **Format:** did the model wrap its thinking in `<think>...</think>` then give `<answer>...</answer>`? A tiny template – the *only* structural constraint imposed.

No neural reward model for reasoning. A learned RM invites *reward hacking* at scale – the policy games the scorer instead of solving the problem. A rule you cannot fool is safer.

The RL algorithm: GRPO (recall from [LLM-1])

R1 optimizes the reward with **GRPO** – the critic-free algorithm from the GRPO deck. We just *recall* it here:

- Sample a **group** of G answers to each question.
- Score each with the rule-based reward.
- Advantage = how far each answer's reward is from the **group mean**, in std units:

$$\hat{A}_i = \frac{r_i - \text{mean}(\mathbf{r})}{\text{std}(\mathbf{r})}$$

- Push up better-than-average answers, push down the rest. **No value network.**

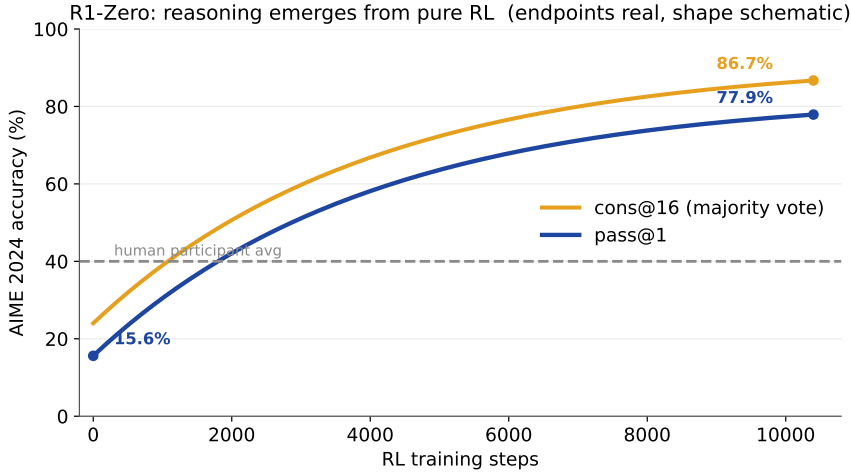
The group mean *is* the baseline – so a reward that only fires on correct answers still gives a usable learning signal: "better than the other $G - 1$ tries."

Config: $G = 16$ samples/question, lr 3×10^{-6}, KL coef 0.001, up to 32k–64k tokens/answer, $\sim$10.4k steps.

R1-Zero: pure RL, no SFT

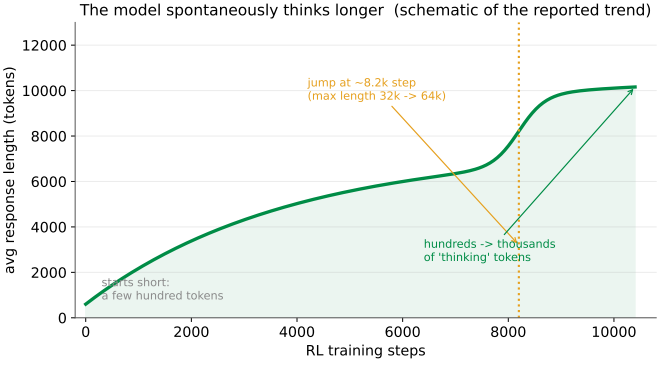
The paper's most striking result – watch reasoning appear on its own.

Accuracy climbs from base toward o1 over RL steps



Pure RL on V3-Base lifts AIME pass@1 from **15.6%** to **77.9%**; majority vote over 16 samples reaches **86.7%**, past the human-participant average – **with no supervised reasoning data at all.**

The model learns to think longer – on its own



Nobody set a target length. As RL proceeds, the model *chooses* to generate more – from a few hundred to thousands of tokens per answer.

Extra tokens are spent **exploring alternatives, re-deriving, and verifying** – the model buys accuracy with test-time thinking.

Curve schematic; the trend and the ∼8.2k-step jump are from the paper's Figure 1.

Predict-first: what behaviors emerge from just a correctness reward?

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It spontaneously learns to **self-verify** ("let me check"), **explore alternative approaches**, and **backtrack** when a path fails – behaviors never demonstrated in any training example.

The "aha moment": mid-training, an intermediate R1-Zero interrupts its own derivation – *"Wait, wait. Wait. That's an aha moment I can flag here. Let's reevaluate this step-by-step..."* – and restarts with a better method. Reflection, learned from reward alone.

But R1-Zero is rough to read

Pure RL bought reasoning power at the cost of **presentation**:

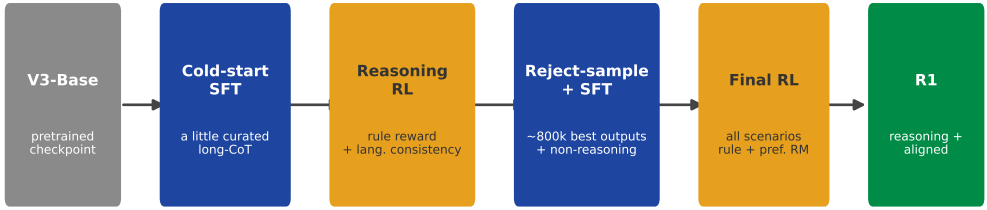
- **Poor readability:** chains are optimized to reach the answer, not to be read – messy, unformatted, hard to follow.
- **Language mixing:** a single chain-of-thought can switch between English and Chinese mid-thought (the base model is bilingual, and nothing penalized the mixing).
- **Narrow:** rule rewards only cover verifiable domains – weak at general writing and open-domain Q&A.

The capability is there; the *package* is not. That gap motivates the full **R1** pipeline – keep the reasoning, fix the delivery.

The full R1 pipeline

Four stages that keep R1-Zero's reasoning but make it readable and general.

Four stages: SFT and RL, alternating



SFT stages (blue) teach readable CoT; RL stages (orange) reward correctness, then broad alignment

A small **cold-start SFT** makes chains readable; **reasoning RL** re-grows the skill (now with a language-consistency reward); **reject-sample + SFT** folds the RL model's own best outputs plus non-reasoning data back in; a **final RL** stage aligns all scenarios.

What each stage does

1. **Cold-start SFT.** Fine-tune V3-Base on a *few thousand* curated long-CoT examples in a clean, conversational format. Just enough to fix readability before RL – *not* enough to teach reasoning.
2. **Reasoning-oriented RL.** GRPO with rule rewards (as R1-Zero), plus a **language-consistency** reward = fraction of the CoT in the target language. Slightly lowers scores but removes the mixing.
3. **Rejection sampling + SFT.** Sample the RL model, *keep the good answers* ($\sim$600k reasoning), add $\sim$200k non-reasoning (writing, QA, code) $\rightarrow$ $\sim$800k examples, and SFT on them.
4. **Final RL for all scenarios.** A last RL pass: rule rewards for reasoning *and* a preference reward model for helpfulness/harmlessness.

The two ideas: **SFT for style, RL for capability** – and **bootstrap** off the model's own best generations.

Distillation

Move R1's reasoning into small dense models – with plain SFT.

Distill R1's reasoning into small dense models

Generate ∼**800k** reasoning examples with R1, then **SFT** open base models on them – *no RL stage for the students*.

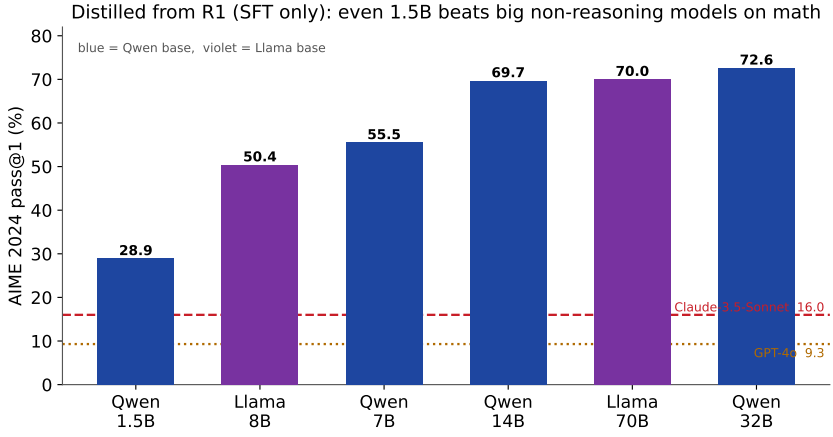
Students: **Qwen** and **Llama** bases, **1.5B** → **70B**.

A pure knowledge-transfer move: the small model learns to imitate R1's long chains-of-thought.

The teacher's reasoning is written down as text. Any model that can be fine-tuned on text can *inherit* it – cheaply, with standard SFT.

2–3 epochs of SFT on the 800k set; the same data used in R1's own stage-3 SFT.

Distilled small models beat big non-reasoning models on math



Even **Distill-Qwen-1.5B** (28.9 on AIME) clears GPT-4o (9.3) and Claude-3.5 (16.0); quality **scales with student size** up to **Distill-Qwen-32B** at 72.6.

Predict-first: distill R1, or run RL on the small model directly?

You have a 32B base. To make it reason, is it better to **distill from R1**, or to run the same **large-scale RL** on the 32B itself?

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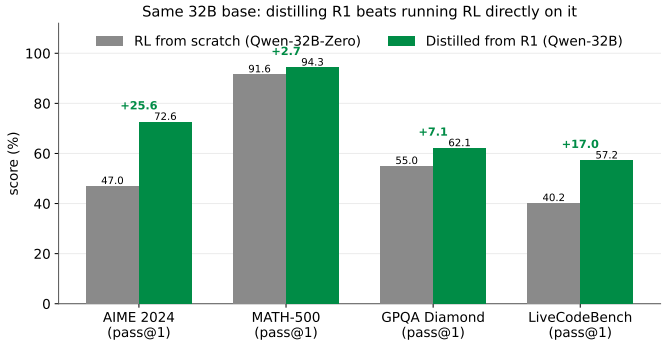
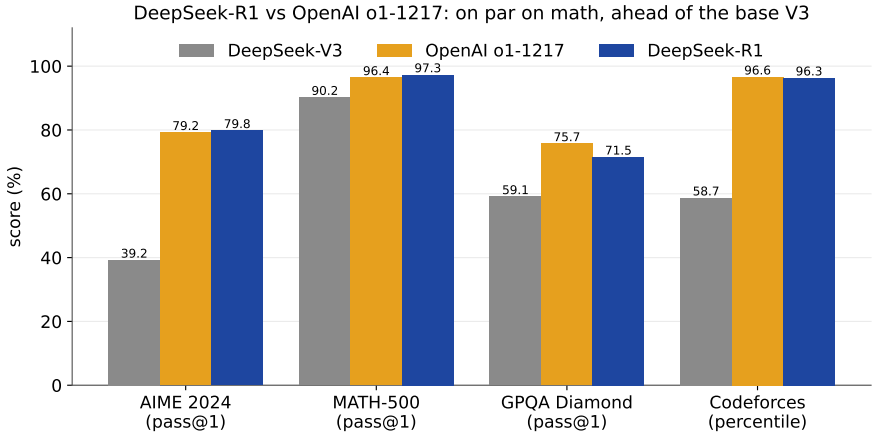


Distillation wins, and not by a little (AIME 72.6 vs 47.0). RL from scratch on a small model needs enormous compute and still trails imitating a stronger teacher's reasoning.

Results and caveats

Where R1 lands against o1 – and where it still falls short.

R1 vs o1-1217, head to head



On math, R1 is **on par** with o1-1217 (AIME 79.8 vs 79.2, MATH-500 97.3 vs 96.4) and far above the base V3. o1 still leads on GPQA; the two trade blows on code.

Caveats and limitations

- **Only where a verifier exists.** Pure RL needs a reliable reward. For open-ended tasks (writing) there is no rule to check, so a learned RM is used – and it can be **reward-hacked**.
- **Language mixing** persists for non-English/Chinese queries.
- **Prompt-sensitive:** few-shot prompting *hurts* R1 – zero-shot, just-state-the-problem works best.
- **No tools** (no calculator/search) and weaker structured output; overthinks easy questions.
- **One report, one lab.** An 86-page technical report; this deck abstracts its headline story, not every ablation.

Still: pure RL from a correctness reward produced frontier reasoning – capability that was *latent in the base model* and **drawn out**, not injected.

Recap

- **R1-Zero:** pure RL (GRPO) on V3-Base with a **rule-based** reward – reasoning **emerges**: longer thinking, self-verification, backtracking, the "aha moment." No SFT, no reasoning traces.
- R1-Zero is capable but **unreadable** and mixes languages.
- **R1 pipeline:** cold-start SFT $\rightarrow$ reasoning RL $\rightarrow$ reject-sample + SFT $\rightarrow$ final RL. *SFT for style, RL for capability.*
- **R1** $\approx$ **o1** on math/code; well ahead of the base V3.
- **Distillation** of R1's chains into 1.5B–70B models **beats** running RL on those small models directly.
- Reward = correctness; the RL **draws out** latent ability rather than injecting new skills.

Next: *[LLM-8] Qwen3* – the same reasoning-RL family, with a single model that can *switch* between slow thinking and fast answering.